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CASE STUDY : 1

Hospital Readmission Records: Data Quality & Wrangling

P24CS904

DATA SCIENCE

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1. Introduction

Healthcare organisations collect large volumes of information during patient visits, including diagnoses, laboratory procedures, medications, length of stay and readmission outcomes. These records can support service planning and quality improvement, but only when the data is sufficiently accurate, consistent and well organised. Missing values, duplicate encounters, inconsistent categories and unusual numeric values can reduce the reliability of analysis.

This case study focuses on preparing the Diabetes 130-US Hospitals dataset for analysis. The work is a data-quality and data-wrangling exercise; it does not attempt to diagnose patients or provide clinical recommendations.

2. Problem Statement

The supplied hospital dataset contains encounter-level records from multiple hospitals. A patient may appear in more than one encounter, and some fields use the character '?' to indicate unavailable information. Diagnosis fields are represented by ICD-9 codes, which are difficult to interpret without a reference table. The task is to profile the dataset, document data-quality issues, apply justified cleaning and transformation operations, integrate diagnosis descriptions, and create a patient-level summary with one row per patient.

3. Objectives

- Understand the structure, dimensions, data types and meaning of the hospital records.
- Identify missing values, duplicate encounter identifiers and inconsistent categories.
- Examine numeric variables for unusual values and document outlier treatment.
- Normalise selected numeric variables for a consistent analytical scale.
- Enrich diagnosis codes by merging a verified ICD-9 lookup table.
- Aggregate encounter-level records into one row per patient.
- Export the cleaned encounter-level and patient-level datasets and maintain a cleaning log.

4. Dataset Description

The Diabetes 130-US Hospitals dataset is available through the UCI Machine Learning Repository. It represents hospital encounters for patients with diabetes across 130 US hospitals over the period 1999–2008. The commonly distributed diabetic_data.csv file contains 101,766 encounter records and 50 columns, including identifiers and the readmission outcome. The repository describes 47 predictive features; the total CSV column count also includes identifier and outcome fields.

Field / Group	Description
patient_nbr	Patient identifier used to link multiple encounters for the same patient.
encounter_id	Unique identifier for a hospital encounter.
race, gender, age	Demographic or patient descriptor fields.
diag_1, diag_2, diag_3	Primary, secondary and additional diagnosis codes.
time_in_hospital	Number of days spent in the hospital for an encounter.
num_lab_procedures, num_procedures	Counts of laboratory and other procedures.
num_medications	Number of medications recorded during an encounter.
readmitted	Readmission outcome category: <30, >30 or NO.

5. Methodology

The workflow is organised into six stages: dataset acquisition, data profiling, cleaning, transformation, ICD-9 enrichment, and patient-level aggregation. The original file should be preserved so that every transformation can be reproduced.

5.1 Data Loading and Profiling

The CSV file is loaded into a Pandas DataFrame. The shape, column names, data types, summary statistics and frequency counts are inspected. The '?' marker is converted to a proper missing-value representation (NaN), allowing Pandas to calculate missing value counts and percentages.

Key profiling operations include df.shape, df.info(), df.describe(include='all'), df.isna().sum(), df.duplicated().sum() and value_counts() for categorical fields.

5.2 Missing Values and Duplicate Records

Missingness is assessed column by column before selecting a treatment. The weight field has a very high proportion of missing values in this dataset and may be excluded from the primary analysis, with the decision recorded. For categorical fields such as race, payer_code and medical_specialty, an explicit 'Unknown' category can preserve records while making missingness visible. Numeric imputation, such as median imputation, should only be applied to fields where it is meaningful and should be documented.

Duplicate rows and duplicate encounter_id values are checked separately. Because encounter_id identifies a hospital encounter, any duplicate identifier should be investigated before removal. A duplicate should not be deleted solely because two different encounters happen to have similar values.

5.3 Outlier Detection and Numeric Normalisation

The Interquartile Range (IQR) method is used to flag potential outliers. For a numeric field, IQR = Q3 − Q1, and the conventional flagging limits are Q1 − 1.5 × IQR and Q3 + 1.5 × IQR. A flagged value is not automatically an error: high numbers of medications, procedures or prior visits may be valid clinical utilisation patterns. Therefore, the project reports outlier counts and retains valid observations unless there is evidence of an invalid value.

Min-Max scaling can transform selected numeric variables to a 0–1 range using x_scaled = (x − x_min) / (x_max − x_min). Identifier columns such as patient_nbr and encounter_id must not be scaled. The unscaled cleaned data should be retained separately so that patient-level totals and averages remain interpretable.

5.4 ICD-9 Lookup Integration

The diagnosis columns diag_1, diag_2 and diag_3 contain diagnosis codes. A separate, verified ICD-9 lookup table is required, with at least a code field and a diagnosis description field. Codes are standardised consistently (for example, by trimming spaces and applying a consistent punctuation convention) before a left join is performed. A left join preserves all hospital encounters. Unmatched codes are reported rather than assigned invented descriptions.

The dataset's IDS_mapping.csv file is not an ICD-9 diagnosis lookup. It maps selected administrative identifiers, including admission type, discharge disposition and admission source. It should only be used for those mappings.

5.5 Patient-Level Reshaping

The encounter data is grouped by patient_nbr using Pandas groupby(). Aggregations can include encounter count, average length of stay, total laboratory procedures, average medications and utilisation counts. The aggregation rule must match the meaning of each field: sums are appropriate for selected encounter counts, while averages or maximums may be more suitable for other measures.

A patient-level readmission indicator must be defined carefully. For example, the maximum of an encounter-level indicator for readmission within 30 days means that at least one of the patient's included encounters had that label. It is not an encounter-specific prediction target and should not be interpreted as the patient's overall clinical status.

6. Cleaning Decisions

Data issue	Proposed treatment	Reason / documentation
'?' missing marker	Convert to NaN	Enables consistent missing-value profiling.
Weight field	Exclude from core analysis if missingness is excessive	Avoid unreliable large-scale imputation; record the exclusion.
Categorical missingness	Use 'Unknown' where appropriate	Preserves records and makes missingness explicit.
Numeric missingness	Impute selectively, e.g. median	Use only where justified; record affected fields and counts.
Duplicate encounter IDs	Investigate and resolve	Preserve distinct encounters; remove only confirmed duplicates.
Potential outliers	Flag with IQR; retain unless invalid	Extreme healthcare utilisation may be legitimate.
Numeric scaling	Min-Max scale selected features	Keep identifiers and a separate unscaled copy.
ICD-9 diagnosis codes	Left-join verified lookup	Preserves encounters and exposes unmatched codes.
Multiple encounters	Group by patient_nbr	Produces one summary row per patient.

7. Implementation Tools

Tool / Library	Purpose
Python	Main programming language for the data-wrangling workflow.
Pandas	Loading, profiling, cleaning, joining, grouping and exporting tabular data.
NumPy	Missing-value representation and numerical operations.
Jupyter Notebook	Interactive execution, explanation and reproducibility.
scikit-learn	MinMaxScaler for numeric normalisation.
Matplotlib / Seaborn	Optional plots for missingness, distributions and outlier inspection.

8. Expected Outputs and Validation

The notebook should generate a cleaned encounter-level CSV, a patient-level CSV, a cleaning-decision log and a reproducible Jupyter Notebook. The final row count and missing-value totals must be taken from the actual program output; they should not be assumed in advance.

Output	Validation check
Cleaned encounter-level dataset	Check row count, column count, missing values and encounter_id uniqueness.
Patient-level dataset	Check that patient_nbr is unique and each patient appears once.
ICD-9-enriched dataset	Count matched and unmatched diagnosis codes for each diagnosis field.
Normalised features	Confirm selected scaled fields fall between 0 and 1; verify identifiers are unchanged.
Cleaning decision log	Confirm each transformation has an issue, action and reason.
Notebook	Restart and run all cells in order to confirm reproducibility.

9. Expected Outcome

The completed workflow is expected to produce a more consistent and analysis-ready representation of the hospital records. Missing-value markers become explicit, categorical values are standardised, potential outliers are documented, selected numeric features are normalised, and diagnosis codes are enriched where a lookup match is available. The patient-level table reduces repeated encounter rows into a single summary per patient.

The result supports descriptive healthcare data analysis and demonstrates core data-wrangling skills. It does not establish that a particular cleaning choice improves patient outcomes, nor does it provide a validated readmission prediction model.

10. Conclusion

This case study demonstrates a structured approach to healthcare data quality and wrangling using the Diabetes 130-US Hospitals dataset. The key lesson is that data cleaning requires transparent, justified decisions rather than blindly deleting records or replacing every unusual value. By profiling the data, documenting missingness, investigating duplicates, flagging outliers, integrating diagnosis descriptions and reshaping encounters to patient-level summaries, the project creates a reproducible foundation for further healthcare analytics. These activities align with SDG 3 by supporting responsible analysis of health-related information.

11. References

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